

The numerical range is a 2-spectral set

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Abstract

For an $n \times n$ complex matrix A , let $W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}$ be its numerical range. Here x^* denotes the conjugate transpose, $\|\cdot\|_2$ the Euclidean norm, and $\|\cdot\|$ the induced operator norm. For every function f holomorphic on a neighborhood of $W(A)$, we prove that

$$\|f(A)\| \leq 2 \max_{z \in W(A)} |f(z)|$$

and consequently that $W(A)$ is a 2-spectral set for A . This proves Crouzeix's conjecture, and the constant is optimal. The central ingredient is a mass-parameterized positive-real completion theorem relative to an auxiliary eigenbasis: if the resolvent defect of a normalized matrix-valued Carathéodory function lies in the adjoint algebra, then its underlying matrix has norm at most the mass $m \geq 2$. The numerical-range double layer has mass two. The classical positive double-layer calculus supplies such completions for $f(B)$ whenever the auxiliary matrix B has simple spectrum; the eigenvalues of $f(B)$ may repeat. Sampling the associated Herglotz kernel at scaled conjugate diagonal entries and at the origin cancels the nonconstant completion term and leaves two ordered weighted Gramians; their first nonconstant terms yield the mass bound. Simple-spectrum approximation and canonical convex outer domains with supported oriented radial boundaries yield the general theorem. The mass formulation also provides a concrete scalar proof protocol for annular, multiply connected, operator-radius, and abstract spectral-constant problems.

Keywords: Crouzeix conjecture; numerical range; spectral set; matrix-valued Herglotz kernel; double-layer potential

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1. Introduction

Let $\mathbb{C}$ denote the complex field and, for $n \geq 1$, let $\mathbb{M}_n(\mathbb{C})$ denote the algebra of $n \times n$ complex matrices. For $A \in \mathbb{M}_n(\mathbb{C})$, its numerical range is

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}. \quad (1)$$

Here x^* denotes the conjugate transpose, $\sigma(A)$ denotes the spectrum, $\|x\|_2$ is the Euclidean norm, $\|X\|$ is the induced operator norm, and scalar moduli are denoted by $|\cdot|$. The set $W(A)$ is compact, the Toeplitz–Hausdorff theorem makes it convex, and $\sigma(A) \subset W(A)$. Functions holomorphic near $\sigma(A)$ are evaluated using the holomorphic functional calculus.

A compact set $K \subset \mathbb{C}$ containing $\sigma(A)$ is called a C -spectral set for A if

$$\|r(A)\| \leq C \max_{z \in K} |r(z)| \quad (2)$$

for every rational function r pole-free on K . Our main result settles Crouzeix's conjecture by identifying the sharp universal constant for $K = W(A)$.

Theorem 1.1 (Main theorem). *Let $n \geq 1$, let $A \in \mathbb{M}_n(\mathbb{C})$, and let f be holomorphic on a neighborhood of $W(A)$. Then*

$$\|f(A)\| \leq 2 \max_{z \in W(A)} |f(z)|. \quad (3)$$

Consequently, $W(A)$ is a 2-spectral set for A .

The factor 2 is optimal: for

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad p(z) = z, \quad (4)$$

one has $\max_{z \in W(J)} |z| = 1/2$ and $\|J\| = 1$.

For later use, write

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

We use I for the identity matrix of the size dictated by context, X^* for the adjoint, put $\operatorname{Re} X = (X + X^*)/2$, and let $\operatorname{alg}(T)$ denote the unital algebra generated by T . For Hermitian matrices X and Y , we write $X \preceq Y$ (equivalently, $Y \succeq X$) when $Y - X$ is positive semidefinite.

Crouzeix proved the estimate with constant 2 for 2×2 matrices and formulated the general conjecture in [1]. His subsequent dimension-free theorem gave the constant 11.08 [2]. After further structural and numerical work [3], Crouzeix and Palencia proved that $W(A)$ is a $(1 + \sqrt{2})$ -spectral set [4]. Their double-layer and Cauchy-transform argument was clarified in [5] and developed in several directions in [6,7].

The intervening literature established the conjectured constant for perturbed Jordan blocks and, more generally, scalar translates of cyclic weighted shifts, as well as for tridiagonal 3×3 matrices with elliptic numerical range centered at an eigenvalue [8–10]. Numerical and extremal investigations appear in [11,12], and a broad account of the problem and its variants is given in [13]. Subsequent work has treated Blaschke-product level sets, cyclicity reductions, compressions of shifts, further nilpotent classes, and quantitative bounds for KMS matrices [14–17]. An alternative proof for a class of weighted shifts appears in [18]. Across these cited reductions, the best dimension-free universal bound remained $1 + \sqrt{2}$.

1.1. The sharpness barrier in the symmetrized calculus

The positive boundary measure used below originates in the convex-domain integral calculus of Delyon–Delyon [19]; see also [4,20]. If B has numerical range in a convex domain Ω with continuously differentiable boundary and f is holomorphic on a neighborhood of $\overline{\Omega}$ with $|f| \leq 1$ on $\overline{\Omega}$, this calculus produces a unital positive map Φ and a holomorphic Cauchy companion g_f such that

$$2\Phi(f) = f(B) + g_f(B)^*. \quad (5)$$

Thus positivity directly controls a symmetrized functional calculus, while the target estimate concerns $f(B)$. The Crouzeix–Palencia argument couples the two terms in (5); the example in [5] shows that its abstract norm estimate is sharp at $1 + \sqrt{2}$. The corresponding general uniform-algebra theorem has the same sharp constant; its additional unital case was proposed as a possible route to 2 in [21]. Schwenninger and de Vries revisited the double-layer method [22]. Malman et al. proved that a configuration-constant refinement gives a strict improvement for every fixed numerical-range shape; they also constructed thin quadrilaterals for which the resulting bounds approach $1 + \sqrt{2}$ [23]. Separately, continuity and compactness give a constant $C_n < 1 + \sqrt{2}$ in every fixed matrix dimension n [24].

The key additional information is algebraic. For the approximants constructed in Section 5, the companion in (5) belongs to the algebra generated by B^* , where the simple-spectrum matrix B supplies

an eigenbasis that also diagonalizes $f(B)$. Treating $f(B)$ and the companion jointly preserves this coupling.

1.2. Auxiliary-basis de-symmetrization

Our numerical-range argument uses the mass-two specialization of the general completion theorem proved below and the whole Cayley family associated with (5). It has four steps.

1. Applying Φ to

$$\zeta \mapsto \frac{1 + wf(\zeta)}{1 - wf(\zeta)}, \quad w \in \mathbb{D},$$

produces a matrix-valued Carathéodory function H , meaning that $H(0) = I$ and $\operatorname{Re} H(w) \succeq 0$.

2. For a simple-spectrum auxiliary matrix B , put $T = f(B)$. The double-layer identity gives the positive-real completion

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*), \quad w \in \mathbb{D}. \quad (6)$$

3. The eigenbasis of B diagonalizes $T = f(B)$ even when some values of f on the spectrum of B coincide, and it turns the unknown term in (6) into a diagonal analytic correction. We sample the Herglotz kernel of H at half the conjugate diagonal entries of T and add one vector sample at the origin. Repeated sampling points are allowed, and the origin sample cancels the diagonal correction's contribution exactly.

4. The surviving positive-matrix inequality compares two weighted Gramians at scales τ and τ^2 , where $\tau = m^{-1}$. Their ordered difference and the first nonconstant Gramian term yield $\|T\| \leq m$; here $m = 2$.

The proof uses two limits: simple-spectrum matrices tend to A , followed by outer domains converging to $W(A)$. Together they yield the general case.

2. Positive-real kernels

Let $\mathcal{K}$ be an $\mathbb{M}_n(\mathbb{C})$ -valued kernel on $\mathbb{D} \times \mathbb{D}$. We call $\mathcal{K}$ positive if, for every finite choice $w_0, \dots, w_m \in \mathbb{D}$ and $\xi_0, \dots, \xi_m \in \mathbb{C}^n$,

$$\sum_{i,j=0}^m \xi_i^* \mathcal{K}(w_i, w_j) \xi_j \geq 0. \quad (7)$$

We shall use the following standard matrix-valued Herglotz kernel fact. A direct proof is included to fix the normalization.

Lemma 2.1 (Matrix Herglotz kernel). *Let $F: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic with $\operatorname{Re} F(w) \succeq 0$ for every $w \in \mathbb{D}$. Then*

$$\mathcal{L}_F(w, z) = \frac{F(w) + F(z)^*}{1 - w\bar{z}} \quad (8)$$

is a positive kernel.

Proof. Fix $w_0, \dots, w_m \in \mathbb{D}$ and $\xi_0, \dots, \xi_m \in \mathbb{C}^n$, and put

$$\eta(\zeta) = \sum_{j=0}^m \frac{\xi_j}{1 - \bar{w}_j \zeta}, \quad |\zeta| = 1.$$

For $0 \leq r < 1$, the scalar Cauchy formula, applied entrywise, gives

$$\frac{1}{2\pi} \int_0^{2\pi} \eta(e^{it})^* 2 \operatorname{Re} F(re^{it}) \eta(e^{it}) dt = \sum_{i,j=0}^m \xi_i^* \frac{F(rw_i) + F(rw_j)^*}{1 - w_i \bar{w}_j} \xi_j. \quad (9)$$

110 The integrand on the left is nonnegative, so the right side is nonnegative. Letting $r \uparrow 1$ and using the
 111 continuity of F at the finitely many points w_i gives (7) for $\mathcal{K} = \mathcal{L}_F$. $\square$

112 3. The positive-real completion principle

113 The following mass-parameterized completion theorem is the core of the argument. It converts a
 114 positive-real completion constrained by the adjoint algebra of an auxiliary simple-spectrum matrix
 115 into a norm bound equal to the mass parameter.

116 **Theorem 3.1** (Positive-real completion of mass m). *Let $2 \leq m < \infty$, and let*

$$117 \quad B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}, \quad T = S \Lambda S^{-1}, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n), \quad (10)$$

118 *where S is invertible, the numbers $\beta_1, \dots, \beta_n$ are distinct, and $|\lambda_i| \leq 1$ for every i . The numbers λ_i may repeat.*
 119 *Let $H: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic and satisfy*

$$120 \quad H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (11)$$

121 *and*

$$122 \quad H(w) - \frac{2}{m}(I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \quad (12)$$

123 *Then $\|T\| \leq m$.*

124 **Proof.** Put $\tau = m^{-1}$, so $0 < \tau \leq 1/2$, and set $G = S^* S \succ 0$. Since the β_i are distinct, polynomial
 125 interpolation shows that

$$126 \quad \operatorname{alg}(B^*) = \left\{ (S^{-1})^* D S^* : D \text{ is diagonal} \right\}. \quad (13)$$

127 By (13), the matrix

$$128 \quad \Theta(w) = S^* (H(w) - 2\tau(I - wT)^{-1}) (S^*)^{-1}$$

129 is diagonal for every w . On setting $\tilde{H}(w) = S^* H(w) S$, we obtain

$$130 \quad \tilde{H}(w) = 2\tau G(I - w\Lambda)^{-1} + \Theta(w)G. \quad (14)$$

131 Since $\tilde{H}(0) = G$, one has $\Theta(0) = (1 - 2\tau)I$. Write

$$132 \quad \Theta(w) = (1 - 2\tau)I + \Psi(w), \quad \Psi(w) = \operatorname{diag}(\psi_1(w), \dots, \psi_n(w)),$$

133 where $\Psi(0) = 0$. Thus

$$134 \quad \tilde{H}(w) = F(w) + \Psi(w)G, \quad F(w) = (1 - 2\tau)G + 2\tau G(I - w\Lambda)^{-1}. \quad (15)$$

135 Since $\operatorname{Re} \tilde{H}(w) = S^* (\operatorname{Re} H(w)) S$, the real part of $\tilde{H}(w)$ is positive semidefinite. Consequently,
 136 Lemma 2.1 applies to $\tilde{H}$.

137 Define Hermitian matrices $P, Q, Y \in \mathbb{M}_n(\mathbb{C})$ by

$$138 \quad P_{ij} = \frac{G_{ij}}{1 - \tau^2 \bar{\lambda}_i \lambda_j}, \quad Q_{ij} = \frac{G_{ij}}{1 - \tau \bar{\lambda}_i \lambda_j}, \quad Y = Q - P. \quad (16)$$

139 The denominators are nonzero because $|\lambda_i| \leq 1$. Since $G = G^*$, conjugate symmetry in the displayed
 140 formulas also gives $P = P^*$ and $Q = Q^*$. Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{C}^n$, and fix $u =$
 141 $(u_1, \dots, u_n)^T \in \mathbb{C}^n$. In the positive-kernel inequality use the n points and vectors

$$142 \quad w_i = \tau \bar{\lambda}_i, \quad \xi_i = u_i e_i \quad (1 \leq i \leq n), \quad (17)$$

143 and add the point $w_0 = 0$ with vector

$$144 \quad \zeta_0 = v = -G^{-1}Pu. \quad (18)$$

145 The points w_i may coincide. Kernel positivity applies to arbitrary finite indexed lists of points and
 146 vectors, including $w_i = w_0$ when $\lambda_i = 0$. We first verify that this sample cancels the contribution of Ψ
 147 to the sampled kernel quadratic form. Its contribution is

$$148 \quad 2 \operatorname{Re} \sum_{i=1}^n \bar{u}_i \psi_i(w_i) \left((Gv)_i + \sum_{j=1}^n \frac{G_{ij}u_j}{1 - w_i \bar{w}_j} \right). \quad (19)$$

149 Because

$$150 \quad 1 - w_i \bar{w}_j = 1 - \tau^2 \bar{\lambda}_i \lambda_j,$$

151 the parenthesis in (19) is $(Gv + Pu)_i$, which vanishes by (18).

152 It remains to calculate the contribution of F . For $x = \bar{\lambda}_r \lambda_s$, the sample-sample entry is

$$153 \quad 2(1 - 2\tau)P_{rs} + \frac{4\tau G_{rs}}{(1 - \tau x)(1 - \tau^2 x)}. \quad (20)$$

154 The partial-fraction identity

$$155 \quad \frac{1}{(1 - \tau x)(1 - \tau^2 x)} = \frac{1}{1 - \tau} \left(\frac{1}{1 - \tau x} - \frac{\tau}{1 - \tau^2 x} \right)$$

156 shows that the compressed sample-sample block is

$$157 \quad M = 2P + \frac{4\tau}{1 - \tau}Y.$$

158 The sample-origin block is $R_0 = 2(1 - \tau)G + 2\tau Q$, and the origin-origin block is $2G$. After cancellation
 159 of the Ψ term, kernel positivity on the graph $v = -G^{-1}Pu$ gives

$$160 \quad 0 \preceq M - R_0 G^{-1}P - P G^{-1}R_0 + 2P G^{-1}P. \quad (21)$$

161 Using $Q = P + Y$ reduces this to

$$162 \quad \frac{4\tau}{1 - \tau}Y - 2\tau(YG^{-1}P + P G^{-1}Y) + 2(1 - 2\tau)(P G^{-1}P - P) \succeq 0. \quad (22)$$

163 Balance the diagonalization by defining

$$164 \quad \begin{aligned} \tilde{T} &= G^{1/2} \Lambda G^{-1/2}, & \hat{P} &= G^{-1/2} P G^{-1/2}, \\ \hat{Q} &= G^{-1/2} Q G^{-1/2}, & \hat{Y} &= \hat{Q} - \hat{P}. \end{aligned} \quad (23)$$

165 Expanding the scalar kernels in (16) into geometric series gives

$$166 \quad \hat{P} = \sum_{k=0}^{\infty} \tau^{2k} \tilde{T}^{*k} \tilde{T}^k, \quad \hat{Q} = \sum_{k=0}^{\infty} \tau^k \tilde{T}^{*k} \tilde{T}^k. \quad (24)$$

167 Indeed, $\tilde{T}^k = G^{1/2} \Lambda^k G^{-1/2}$, so the sequence $(\tilde{T}^k)$ is bounded even when some $|\lambda_j| = 1$; the geometric
 168 weights give norm convergence. It follows that

$$169 \quad \hat{P} \succeq I, \quad \hat{Y} = \sum_{k=1}^{\infty} (\tau^k - \tau^{2k}) \tilde{T}^{*k} \tilde{T}^k \succeq \frac{1 - \tau}{\tau} (\hat{P} - I). \quad (25)$$

The last inequality follows termwise. Taking the congruence of (22) by $G^{-1/2}$ and dividing by 2τ yields

$$\frac{2}{1-\tau}\hat{Y} - \hat{Y}\hat{P} - \hat{P}\hat{Y} + \frac{1-2\tau}{\tau}(\hat{P}^2 - \hat{P}) \succeq 0. \quad (26)$$

Let e be a unit eigenvector of $\hat{P}$, say $\hat{P}e = \alpha e$, and put $y = e^*\hat{Y}e$. Then $\alpha \geq 1$ and (26) gives

$$0 \leq 2\left(\frac{1}{1-\tau} - \alpha\right)y + \frac{1-2\tau}{\tau}\alpha(\alpha-1). \quad (27)$$

If $\alpha \leq (1-\tau)^{-1}$, then $\alpha \leq 2$. Otherwise the coefficient of y is negative, while $y \geq (1-\tau)(\alpha-1)/\tau$ by (25). Because the coefficient is negative, this lower bound for y gives an upper bound for the right side of (27), namely

$$0 \leq \frac{\alpha-1}{\tau}(2-\alpha).$$

Thus $\alpha \leq 2$ in either case. Therefore

$$I \preceq \hat{P} \preceq 2I. \quad (28)$$

The $k = 1$ term of the same series now gives

$$\tau^2 \tilde{T}^* \tilde{T} \preceq \hat{P} - I \preceq I.$$

Thus $\|\tilde{T}\| \leq \tau^{-1} = m$. The polar decomposition $S = UG^{1/2}$ gives

$$T = U\tilde{T}U^*,$$

where U is unitary. Therefore $\|T\| \leq m$. $\square$

Remark 3.2. The sampling scale $\tau = m^{-1}$ simultaneously produces the weights τ and τ^2 in (24). Their ordered difference bounds $\hat{P}$, and the first nonconstant term then gives the coefficient m . The origin sample cancels the nonconstant diagonal correction exactly. When $m = 2$, the last term of (26) vanishes and the theorem reduces to the sharp completion used for numerical ranges. For $m > 2$, the nonzero constant part $\Theta(0) = (1 - 2/m)I$ produces that additional positive Gramian term; omitting it would invalidate the parameterized proof. Distinct auxiliary eigenvalues identify $\text{alg}(B^*)$ with the diagonal algebra in the chosen basis, while the diagonal entries of T may repeat.

4. The double-layer completion

We now construct from the numerical range the mass-two function required by Theorem 3.1. The measure and companion transform used here are classical [2,4,19]. The full Cayley family yields the algebraic constraint (12) relative to the auxiliary matrix.

Only the canonical parallel domains used in Section 5 are needed. We say that a bounded convex domain Ω has a *supported oriented radial boundary* if, for some $c \in \Omega$, its boundary is traced once by

$$\gamma(t) = c + \rho(t)e^{it}, \quad 0 \leq t \leq 2\pi,$$

where ρ is positive, continuously differentiable, and 2π -periodic, and there are continuous 2π -periodic functions v and s such that $|v(t)| = 1$, $s(t) \geq 0$,

$$\gamma'(t) = iv(t)s(t), \quad \text{Re}(\overline{v(t)}(\gamma(t) - z)) \geq 0 \quad (z \in \Omega). \quad (29)$$

Thus v is the outward unit normal, $s(t) dt$ is arclength, and the orientation is positive. Lemma 5.2 below supplies exactly this boundary for every canonical parallel domain.

Lemma 4.1 (Radial double-layer positive map). *Let $B \in \mathbb{M}_n(\mathbb{C})$, let $\Omega \subset \mathbb{C}$ be a bounded convex domain with a supported oriented radial boundary γ , and suppose $W(B) \subset \Omega$. There is a bounded complex-linear, unital, positive map on the compact parameter interval,*

$$\Phi_\gamma: C([0, 2\pi]) \longrightarrow \mathbb{M}_n(\mathbb{C}),$$

given by the double-layer density below.

Proof. Use the data in (29), and put

$$Z_t = \gamma(t)I - B, \quad \mathcal{R}_t = Z_t^{-1}.$$

This resolvent exists because $\sigma(B) \subset W(B) \subset \Omega$. Define the matrix-valued boundary density

$$D_B(t) = \frac{s(t)}{2\pi} \left(v(t)\mathcal{R}_t + \overline{v(t)}\mathcal{R}_t^* \right). \quad (30)$$

It is positive semidefinite. The support inequality in (29) gives

$$v(t)Z_t^* + \overline{v(t)}Z_t = 2\operatorname{Re}(\overline{v(t)}Z_t) \succeq 0. \quad (31)$$

Indeed, for every unit vector $x \in \mathbb{C}^n$, the quadratic form on the left is

$$2\operatorname{Re}(\overline{v(t)}(\gamma(t) - x^*Bx)) \geq 0,$$

because $x^*Bx \in W(B) \subset \Omega$. Congruence by $\mathcal{R}_t$ turns the left side of (31) into the matrix in parentheses in (30).

Since $d\gamma = iv(t)s(t)dt$, the matrix Cauchy formula gives

$$\frac{1}{2\pi} \int_0^{2\pi} v(t)\mathcal{R}_t s(t) dt = I. \quad (32)$$

Taking adjoints gives the same identity for the second layer. Hence

$$\int_0^{2\pi} D_B(t) dt = 2I. \quad (33)$$

Thus the raw positive layer has mass $2I$. For $\varphi \in C([0, 2\pi])$, define

$$\Phi_\gamma(\varphi) = \frac{1}{2} \int_0^{2\pi} \varphi(t) D_B(t) dt. \quad (34)$$

The factor $1/2$ normalizes this mass. The positivity of D_B and (33) make Φ_γ positive and unital, hence $*$ -preserving. It is bounded because

$$\|\Phi_\gamma(\varphi)\| \leq \frac{1}{2} \|\varphi\|_\infty \int_0^{2\pi} \|D_B(t)\| dt.$$

□

Proposition 4.2 (Radial Cayley completion). *Let Ω be a bounded convex domain with a supported oriented radial boundary, let $B \in \mathbb{M}_n(\mathbb{C})$ have simple spectrum, and suppose $W(B) \subset \Omega$. Let V be an open neighborhood of $\overline{\Omega}$, and let f be holomorphic on V and satisfy*

$$\max_{\zeta \in \overline{\Omega}} |f(\zeta)| \leq 1. \quad (35)$$

233 Put $T = f(B)$. Then there is an analytic $H: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ satisfying

$$234 \quad H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (36)$$

235 and

$$236 \quad H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \quad (37)$$

237 **Proof.** Choose

$$238 \quad B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}.$$

239 Then $T = S \operatorname{diag}(f(\beta_1), \dots, f(\beta_n)) S^{-1}$, and (35) gives $|f(\beta_i)| \leq 1$ for every i . In particular, $\sigma(T) \subset \overline{\mathbb{D}}$.

240 For the fixed boundary data, put $\mathcal{R}_t = (\gamma(t)I - B)^{-1}$, and let Φ_γ be the bounded map from
241 Lemma 4.1. For $w \in \mathbb{D}$, define

$$242 \quad c_w(t) = \frac{1 + wf(\gamma(t))}{1 - wf(\gamma(t))} \quad (38)$$

243 and put

$$244 \quad H(w) = \Phi_\gamma(c_w). \quad (39)$$

245 The expansion

$$246 \quad c_w(t) = 1 + 2 \sum_{m=1}^{\infty} w^m f(\gamma(t))^m \quad (40)$$

247 converges locally uniformly in $C([0, 2\pi])$. The boundedness of Φ_γ therefore makes H analytic, and
248 $H(0) = I$. Furthermore,

$$249 \quad \operatorname{Re} c_w(t) = \frac{1 - |wf(\gamma(t))|^2}{|1 - wf(\gamma(t))|^2} \geq 0.$$

250 The positivity and $*$ -preservation of Φ_γ give $\operatorname{Re} H(w) = \Phi_\gamma(\operatorname{Re} c_w) \succeq 0$.

251 For every $m \geq 0$, the scalar Cauchy formula applied in the chosen eigenbasis gives exactly

$$252 \quad \frac{1}{2\pi} \int_0^{2\pi} f(\gamma(t))^m v(t) \mathcal{R}_t s(t) dt = T^m. \quad (41)$$

253 Indeed, this is the diagonal collection of the scalar Cauchy formulas for f^m at the points β_i ; the
254 neighborhood V contains the contour and its interior. Transporting the uniformly convergent series (40)
255 through this first-layer integral gives

$$256 \quad I + 2 \sum_{m=1}^{\infty} w^m T^m = (I + wT)(I - wT)^{-1}.$$

257 The matrix series converges because the powers of T are bounded in the displayed eigenbasis.

258 Define the companion matrix

$$259 \quad G(w) = \frac{1}{2\pi} \int_0^{2\pi} \overline{c_w(t)} v(t) \mathcal{R}_t s(t) dt.$$

260 Splitting the density (30) into its two layers now gives

$$261 \quad 2H(w) = (I + wT)(I - wT)^{-1} + G(w)^*.$$

262 Each $\mathcal{R}_t$ belongs to $\operatorname{alg}(B)$, and this finite-dimensional algebra is closed, so $G(w) \in \operatorname{alg}(B)$. Since
263 $(I + wT)(I - wT)^{-1} = 2(I - wT)^{-1} - I$, we conclude that

$$264 \quad H(w) - (I - wT)^{-1} = \frac{1}{2} (G(w)^* - I) \in \operatorname{alg}(B^*). \quad (42)$$

265 This proves (37). $\square$

Corollary 4.3 (Auxiliary sharp bound). *Under the hypotheses of Proposition 4.2, $\|f(B)\| \leq 2$.*

Proof. Write

$$B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}.$$

Then

$$T = f(B) = S \operatorname{diag}(f(\beta_1), \dots, f(\beta_n)) S^{-1}.$$

Since $\sigma(B) \subset \Omega$, the bound on f gives $|f(\beta_i)| \leq 1$ for every i . Apply Theorem 3.1 with $m = 2$ to these simultaneous diagonalizations and to the function H constructed in Proposition 4.2. The values $f(\beta_i)$ may repeat. $\square$

5. Passage to arbitrary matrices and proof of the main theorem

Successive approximations pass from the auxiliary setting to arbitrary matrices.

Lemma 5.1 (Simple spectrum and numerical range). *Matrices with n distinct eigenvalues are dense in $\mathbb{M}_n(\mathbb{C})$. Moreover, for $A, B \in \mathbb{M}_n(\mathbb{C})$,*

$$W(B) \subset W(A) + \{z \in \mathbb{C} : |z| \leq \|B - A\|\}. \quad (43)$$

Proof. The discriminant of the characteristic polynomial is a nonzero polynomial in the matrix entries, as is seen at $\operatorname{diag}(1, \dots, n)$. The zero set of a nonzero complex polynomial has empty interior, so its nonvanishing set is dense in $\mathbb{M}_n(\mathbb{C})$, proving the first assertion. For a unit vector x ,

$$|x^*(B - A)x| \leq \|B - A\|,$$

which proves (43). $\square$

Lemma 5.2 (Parallel convex domains). *Let $K \subset \mathbb{C}$ be nonempty, compact, and convex. For $\delta > 0$, the set*

$$\Omega_\delta = \{z \in \mathbb{C} : \operatorname{dist}(z, K) < \delta\}$$

is a bounded convex domain, $\overline{\Omega}_\delta = K + \delta\overline{\mathbb{D}}$, and Ω_δ has a supported oriented radial boundary.

Proof. The set Ω_δ is bounded and convex, and its closure is the Minkowski sum $K + \delta\overline{\mathbb{D}}$. Identify $\mathbb{C}$ with $\mathbb{R}^2$. The metric projection $\pi_K: \mathbb{C} \rightarrow K$ is unique and continuous. Put $d_K(z) = \operatorname{dist}(z, K)$. For $z, h \in \mathbb{C}$, comparison first with $\pi_K(z)$ and then with $\pi_K(z + h)$ gives

$$\begin{aligned} 2 \operatorname{Re}(\overline{z + h - \pi_K(z + h)} h) - |h|^2 &\leq d_K(z + h)^2 - d_K(z)^2 \\ &\leq 2 \operatorname{Re}(\overline{z - \pi_K(z)} h) + |h|^2. \end{aligned}$$

The continuity of π_K now shows that d_K^2 is continuously differentiable, with real gradient $2(z - \pi_K(z))$. On the level set $d_K(z) = \delta$ this gradient has norm $2\delta > 0$. Since $\partial\Omega_\delta = \{d_K = \delta\}$, the implicit-function theorem shows that $\partial\Omega_\delta$ is continuously differentiable.

Choose $c \in K$. The ball $c + \delta\mathbb{D}$ lies in Ω_δ , so every ray from c meets the boundary once and transversely. Its radial function ρ is therefore positive, continuously differentiable, and periodic. With

$$\nu(t) = \frac{\gamma(t) - \pi_K(\gamma(t))}{\delta} \quad (0 \leq t \leq 2\pi)$$

as the outward unit normal along γ and $s(t) = |\gamma'(t)|$, convexity gives the support inequality in (29); positive orientation gives $\gamma'(t) = i\nu(t)s(t)$. These data form the required supported oriented radial boundary. $\square$

Proof of Theorem 1.1. Put $K = W(A)$, and let U be an open neighborhood of K on which f is holomorphic. Choose $\varepsilon_0 > 0$ such that the closed ε_0 -neighborhood of K lies in U . For $0 < \varepsilon < \varepsilon_0$, let

$$\Omega_\varepsilon = \{z \in \mathbb{C} : \text{dist}(z, K) < \varepsilon\}.$$

Lemma 5.2 makes this a bounded convex domain with a supported oriented radial boundary. The open convex set $V = \Omega_{\varepsilon_0}$ satisfies $\overline{\Omega_\varepsilon} \subset V \subset U$.

By Lemma 5.1, choose simple-spectrum matrices $B_k \rightarrow A$. For fixed ε , all sufficiently large k satisfy $\|B_k - A\| < \varepsilon/2$, and then (43) gives

$$W(B_k) \subset K + (\varepsilon/2)\overline{\mathbb{D}} \subset \Omega_\varepsilon.$$

Set

$$m_\varepsilon = \max_{z \in \overline{\Omega_\varepsilon}} |f(z)|.$$

If $m_\varepsilon = 0$, then f vanishes on a neighborhood of $\sigma(A)$, so $f(A) = 0$ and there is nothing to prove. Otherwise, apply Corollary 4.3 to B_k and f/m_ε to obtain

$$\|f(B_k)\| \leq 2m_\varepsilon.$$

Continuity of the holomorphic functional calculus, obtained from a fixed contour inside U , gives $f(B_k) \rightarrow f(A)$. Hence

$$\|f(A)\| \leq 2m_\varepsilon. \quad (44)$$

Put $m_0 = \max_{z \in K} |f(z)|$. Uniform continuity of f on the closed ε_0 -neighborhood of K gives a modulus of continuity ω_f there. Since every point of $\overline{\Omega_\varepsilon}$ lies within ε of K ,

$$0 \leq m_\varepsilon - m_0 \leq \omega_f(\varepsilon).$$

Letting $\varepsilon \downarrow 0$ in (44) proves (3).

Finally, a rational function pole-free on the compact set $W(A)$ is holomorphic on a neighborhood of $W(A)$. Applying (3) to such functions proves (2) with $K = W(A)$ and $C = 2$. $\square$

6. Consequences and scope

The Hilbert-space extension follows by the standard finite-dimensional compression argument from [2]; compare the survey [13].

Corollary 6.1 (Hilbert-space form). *Let $\mathcal{H} \neq \{0\}$ be a complex Hilbert space, with inner product linear in the first variable, and let $\mathcal{B}(\mathcal{H})$ denote its algebra of bounded operators. For $A \in \mathcal{B}(\mathcal{H})$, put*

$$W(A) = \{\langle Ax, x \rangle : x \in \mathcal{H}, \|x\| = 1\}.$$

Then, for every $p \in \mathbb{C}[z]$,

$$\|p(A)\| \leq 2 \sup_{z \in W(A)} |p(z)|. \quad (45)$$

Consequently, $\overline{W(A)}$ is a 2-spectral set for A .

Proof. The assertion is immediate if $p = 0$. Otherwise, write $d = \deg p$. For a unit vector $x \in \mathcal{H}$, let

$$\mathcal{M}_x = \text{span}\{x, Ax, \dots, A^d x\}, \quad A_x = \Pi_x A|_{\mathcal{M}_x},$$

where Π_x is the orthogonal projection onto $\mathcal{M}_x$. Induction gives

$$A_x^k x = A^k x \quad (0 \leq k \leq d), \quad (46)$$

because $A^{k+1}x \in \mathcal{M}_x$ whenever $k < d$. Moreover, $W(A_x) \subset W(A)$. Theorem 1.1, applied on the finite-dimensional space $\mathcal{M}_x$, therefore gives

$$\|p(A)x\|_{\mathcal{H}} = \|p(A_x)x\|_{\mathcal{H}} \leq \|p(A_x)\| \leq 2 \sup_{z \in W(A)} |p(z)|.$$

Taking the supremum over unit vectors x proves (45).

Put $K = \overline{W(A)}$. The standard inclusion $\sigma(A) \subset K$ will also be used below. Let $r = a/b$ be a reduced rational function with no pole on K . Then b has no zero on K , so spectral mapping and $\sigma(A) \subset K$ make $b(A)$ invertible.

The complement of the compact convex set K is connected. Polynomial Runge approximation therefore gives polynomials p_j converging uniformly to r on K . By (45),

$$\|p_j(A) - p_k(A)\| \leq 2 \max_{z \in K} |p_j(z) - p_k(z)|,$$

so $p_j(A)$ converges in operator norm to some R , with

$$\|R\| \leq 2 \max_{z \in K} |r(z)|.$$

Since $p_j b \rightarrow a$ uniformly on K , the same polynomial estimate gives

$$p_j(A)b(A) = (p_j b)(A) \rightarrow a(A).$$

Thus $Rb(A) = a(A)$ and $R = a(A)b(A)^{-1} = r(A)$. This proves the rational estimate and hence the spectral-set assertion. $\square$

The conclusion also controls polynomial and rational approximation of matrix functions. Let $A \in \mathbb{M}_n(\mathbb{C})$, let f be holomorphic on a neighborhood of $W(A)$, and let s be either a polynomial or a rational function pole-free on $W(A)$. Then $f - s$ is holomorphic on a neighborhood of $W(A)$, and Theorem 1.1 gives

$$\|f(A) - s(A)\| \leq 2 \max_{z \in W(A)} |f(z) - s(z)|. \quad (47)$$

This is the sharp universal form of the numerical-range estimate that underlies applications to GMRES and rational Krylov methods; see [7,25,26] for the role of spectral sets in those settings.

7. Discussion and further directions

The proof separates an algebraic completion engine from its geometric input. This makes it possible to say precisely when another scalar spectral-constant problem is ready for the same method.

7.1. A reusable scalar protocol

Let $\mathcal{A}$ be a uniform algebra on a compact space K , let $\theta: \mathcal{A} \rightarrow \mathbb{M}_n(\mathbb{C})$ be a continuous unital homomorphism, and suppose that there is a simple-spectrum matrix B such that $\theta(\mathcal{A}) \subset \text{alg}(B)$. The following criterion is the functional-calculus form of Theorem 3.1.

Proposition 7.1 (Positive-layer transfer). *Suppose $2 \leq m < \infty$ and that a bounded complex-linear map $\mathcal{P}: \mathcal{A} \rightarrow \mathbb{M}_n(\mathbb{C})$ satisfies*

$$\mathcal{P}(1) = mI, \quad \text{Re } h \geq 0 \implies \text{Re } \mathcal{P}(h) \succeq 0, \quad \mathcal{P}(h) - \theta(h) \in \text{alg}(B^*) \quad (48)$$

for every $h \in \mathcal{A}$. Then $\|\theta\| \leq \mathfrak{m}$.

Proof. Fix $f \in \mathcal{A}$ with $\|f\|_K \leq 1$, put $T = \theta(f)$, and, for $w \in \mathbb{D}$, define

$$c_w = (1 + wf)(1 - wf)^{-1}, \quad H(w) = \mathfrak{m}^{-1} \mathcal{P}(c_w).$$

The diagonal entries of T are contractive character values, while c_w has nonnegative real part. Hence H is analytic, $H(0) = I$, and $\operatorname{Re} H(w) \succeq 0$. Since

$$\theta(c_w) = (I + wT)(I - wT)^{-1} = 2(I - wT)^{-1} - I,$$

condition (48) gives

$$H(w) - \frac{2}{\mathfrak{m}}(I - wT)^{-1} = \frac{1}{\mathfrak{m}}(\mathcal{P}(c_w) - \theta(c_w) - I) \in \operatorname{alg}(B^*).$$

Theorem 3.1 yields $\|\theta(f)\| \leq \mathfrak{m}$; take the supremum over the unit ball of $\mathcal{A}$. $\square$

Thus a finite-dimensional scalar problem is ready once four items are available: a positive layer, its total mass, an adjoint-algebra defect, and an approximation by simultaneous simple-spectrum models. The upper bound produced by the method is the layer mass, not the number of boundary components by itself. Proposition 7.1 also resolves the finite-dimensional semisimple-range case of the abstract Crouzeix problem in [21, Conjecture 1.2]: if α is the unital antilinear contraction in that conjecture and

$$\theta_\alpha(h) = \frac{1}{2}(\theta(h) + \theta(\alpha(h))^*)$$

is contractive, then $\mathcal{P} = 2\theta_\alpha$ has mass two. Indeed, each map $h \mapsto x^* \theta_\alpha(h) x$, for $\|x\| = 1$, is a unital functional of norm one, hence extends to a state on $C(K)$ and preserves nonnegative real parts. Moreover, $\mathcal{P}(h) - \theta(h) = \theta(\alpha(h))^*$. Simultaneous diagonalization of the semisimple range, followed by the choice of an auxiliary simple diagonal matrix, verifies the remaining hypothesis.

7.2. Formal scalar consequences

Several consequences require no new boundary layer. A faithful universal representation of a unital C^* -algebra $\mathfrak{C}$ transfers Corollary 6.1 to its algebraic numerical range $V_{\mathfrak{C}}(a)$:

$$\|f(a)\| \leq 2 \max_{z \in V_{\mathfrak{C}}(a)} |f(z)|. \quad (49)$$

Here f may be any rational function with poles off $V_{\mathfrak{C}}(a)$. Applying this to the Calkin algebra gives

$$\|f([A])\| \leq 2 \max_{z \in W_e(A)} |f(z)|$$

for every rational f with poles off $W_e(A)$, where $[A]$ is the Calkin coset. If f is also pole-free on $\sigma(A)$, the left side equals $\|f(A)\|_e$. For separable infinite-dimensional Hilbert spaces, [27, Theorem 5.2] identifies the universal Calkin constant with the universal Hilbert-space constant, and hence yields the same conclusion once Corollary 6.1 is used.

For a closed densely defined operator satisfying $\sigma(A) \subset W(A)$ and $\overline{W(A)} \subset \Omega$, where $\Omega \neq \mathbb{C}$ is an open unbounded convex domain of aperture 2α , $0 \leq \alpha \leq \pi/2$, the Yosida approximation in [28, Section 3] combines with Corollary 6.1 to give, for every rational f bounded on Ω and with poles off $\sigma(A)$,

$$\|f(A)\| \leq \min \left\{ 2, 1 - \frac{\alpha}{\pi} + \sqrt{2 - \frac{4\alpha}{\pi} + \frac{\alpha^2}{\pi^2}} \right\} \sup_{z \in \Omega} |f(z)|. \quad (50)$$

The value 2 improves the aperture estimate exactly when $0 \leq \alpha < \pi/6$.

Operator-radius constants transfer formally as well. The conversion theorem of Badea, Crouzeix, and Klaja gives, from a sharp scalar norm constant two,

$$H_\rho(2) = \frac{5 + \sqrt{25 - 16\rho(2 - \rho)}}{4\rho}, \quad \rho \geq 1; \quad H_2(2) = \frac{5}{4}. \quad (51)$$

See [29, Theorems 3.2 and 5.1].

7.3. New mass-two layers and short geometric transfers

The quantum annulus is the cleanest genuinely new positive-layer test. Fix $R > 1$ and orient the two circles of $\mathbb{A}_R = \{R^{-1} < |z| < R\}$ positively. If B is invertible with $\|B\|, \|B^{-1}\| < R$, subtract from the annular double-layer density

$$\mu_B(z) = \frac{1}{2\pi i} \left[z'(s)(zI - B)^{-1} - \overline{z'(s)}(\bar{z}I - B^*)^{-1} \right]$$

the scalar winding density

$$\mathcal{D}_B^{\text{ann}}(z) = \mu_B(z) - \frac{1}{2\pi i} \frac{z'(s)}{z} I. \quad (52)$$

On the outer circle and on the clockwise inner circle, with $r = R^{-1}$, direct multiplication gives respectively

$$\begin{aligned} 2\pi R \mathcal{D}_B^{\text{ann}}(z) &= (zI - B)^{-1} (R^2 I - BB^*) (\bar{z}I - B^*)^{-1} \succeq 0, \\ 2\pi r \mathcal{D}_B^{\text{ann}}(z) &= (zI - B)^{-1} (BB^* - r^2 I) (\bar{z}I - B^*)^{-1} \succeq 0. \end{aligned} \quad (53)$$

The scalar winding numbers cancel, so this layer has mass $2I$; the annular Cauchy identity leaves an adjoint Cauchy companion in $\text{alg}(B^*)$. If B has simple spectrum, all hypotheses of Proposition 7.1 are therefore present. Arbitrary matrices in the strict class follow by simple-spectrum approximation, and radius enlargement gives the non-strict boundary case. This yields the finite-dimensional upper constant two. Tsikalas's bilateral weighted shift gives the matching lower bound for the universal all-Hilbert-space constant [30]; it does not by itself prove finite-dimensional sharpness. For context, the previous all-Hilbert-space upper bound was $1 + \sqrt{2}$ [7], and a more recent matrix bound is $1 + \sqrt{1 + 2/(R^2 + 1)}$ [31]. The boundary dilation of McCullough and Pascoe, together with residual finite dimensionality of the relevant free product, can be combined in a companion argument to pass from matrices to arbitrary Hilbert spaces [32, Theorem 1.1 and Proposition 3.2], [33, Theorems 2.4 and 3.2].

The centered annular classes of Jury and Tsikalas provide a second ready input. Their contractive double-layer calculus has mass two when $c = 0$ and mass $2 + c$ in general; compare their Definition 3.6 and Theorem 3.12 in [34]. Theorem 3.12 assumes that the spectrum lies in the open annulus; the closed-boundary class requires radius enlargement followed by a stable limit. Subject to that reduction, at $c = 0$ the mass-two theorem supplies the finite-dimensional upper constant two, improving their general $1 + \sqrt{2}$ estimate [34, Theorem 5.3] and extending their special 2×2 constant-two result [34, Theorem 1.3]. For $c > 0$, in finite dimensions, Theorem 3.1 supplies the upper coefficient $2 + c$, without a sharpness assertion.

Other problems invite a short transfer from the numerical-range theorem rather than a new layer. For $A \in \mathbb{M}_n(\mathbb{C})$ and $0 < |q| \leq 1$, put

$$W_q(A) = \{ \langle Ax, y \rangle : \|x\| = \|y\| = 1, \langle x, y \rangle = q \}, \quad \Omega_q = q^{-1} W_q(A).$$

442 Tsing proves the convexity of $W_q(A)$ [35], while O’Loughlin and Rani propose in [36, Conjecture 4.2]
 443 the estimate $\|p(A)\| \leq C_q \sup_{z \in \Omega_q} |p(z)|$, with the candidate

$$444 \quad C_q = \max \left\{ 1, \frac{2|q|}{1 + \sqrt{1 - |q|^2}} \right\}. \quad (54)$$

445 Deriving this value from Theorem 1.1 requires a separate rank-one-similarity and scalar-extraction
 446 argument; that step is not contained in the two cited sources. Likewise, every ellipse containing $W(A)$
 447 has upper constant two immediately; fixed-ellipse sharpness and the Douglas–Paulsen norm require
 448 the structured ellipse and annular models in [37,38]. The two-disk setup in [7, Theorem 10] assumes
 449 $\|A - w_1 I\| \leq R_1$ and $\|(A - w_2 I)^{-1}\| \leq R_2$, and distinguishes nested, crossing, and tangent boundary
 450 circles. For this setup the crossing case admits a Möbius reduction to a convex disk–half-plane
 451 geometry, while a genuine nested annular case reduces to the quantum annulus.

452 7.4. Higher-mass scalar problems

453 The mass theorem also identifies several finite-dimensional problems whose algebraic part is
 454 complete and whose remaining work is boundary factorization and approximation.

- 455 1. For a disk with N disjoint circular holes, subtracting one unit of mass on the outer circle and
 456 adding one on each inner circle leaves a positive layer of mass $N + 1$. The resulting target bound
 457 is $N + 1$. The underlying class originates in [39]. A recent geometry-independent bound is
 458 $(N + 1)/2 + \sqrt{(N + 1)^2/4 + N}$, and the question whether $N + 1$ is a universal lower bound
 459 appears in [40, Theorem 2.3 and Question 2.5]. Sharpness remains open for $N \geq 2$.
- 460 2. For the numerical annulus $w(A) \leq R, w(A^{-1}) \leq R$, the outer convex layer has mass two and the
 461 shifted inner layer adds two, producing the target upper bound four. The universal all-Hilbert-
 462 space lower bound is two by inclusion of the quantum-annulus class [30], while the recent matrix
 463 estimate is $2 + \sqrt{4 + 2/(R^2 + 1)}$ [31]; the sharp finite-dimensional constant remains open.
- 464 3. For $\text{DL}\mathbb{A}_R(c)$ with $c > 0$, the natural mass is $2 + c$, as above. Its optimality is not implied by the
 465 completion theorem.

466 Polyannuli and intersections of three or more arbitrary spherical disks are not automatic consequences:
 467 they introduce several kernels or fail to reduce to one annulus or one convex numerical-range problem.
 468 Nor do the finite-dimensional higher-mass arguments pass to arbitrary Hilbert spaces without a
 469 separate stable reduction.

470 7.5. The matrix-valued obstruction and extremizers

471 The principal strengthening still outside the method is the uniform matrix-polynomial question:
 472 for every $m \geq 1$ and every $\mathcal{F} = [f_{ij}] \in \mathbb{M}_m(\mathbb{C}[z])$, does one have

$$473 \quad \|\mathcal{F}(A)\| \leq 2 \max_{z \in W(A)} \|\mathcal{F}(z)\|, \quad \mathcal{F}(A) = [f_{ij}(A)]? \quad (55)$$

474 The case $m = 1$ is Theorem 1.1; uniformity in m is an additional operator-algebraic requirement. At
 475 matrix level the correction occupies $\text{alg}(B^*) \otimes \mathbb{M}_m$, so its diagonal entries are operator blocks and the
 476 scalar cancellation in (19) no longer yields the Gramian inequality. The foundational disk similarity
 477 estimate is due to Okubo and Ando [41]; see also [42,43]. The complete $1 + \sqrt{2}$ bound follows from [4],
 478 and complete constant two is known for the weighted-shift family [18]. Thus no complete constant-two,
 479 similarity, or normal-dilation conclusion is asserted here.

480 Finally, specialize the proof of Theorem 3.1 to mass two. If $\|\tilde{T}\| = 2$ and a unit vector e satisfies
 481 $\|\tilde{T}e\|_2 = 2$, then (28) and (24) imply

$$482 \quad \hat{P}e = 2e, \quad \tilde{T}^2e = 0, \quad \hat{P}\tilde{T}e = \tilde{T}e.$$

These saturation conditions may classify extremal pairs and quantify stability near (4); any classification must allow the nonuniqueness phenomena in [12].

AI-assistance disclosure

OpenAI ChatGPT contributed the idea of sampling the matrix Herglotz kernel at scaled conjugate eigenvalues, with scale $1/2$ in the mass-two application, and adding the origin sample that cancels the nonconstant adjoint-algebra correction's contribution. This appears in the proof of Theorem 3.1, especially (17)–(19). It also assisted with exposition, bibliography, and typesetting. The human author assumes full responsibility for the correctness of all mathematical statements and for the integrity and accuracy of all citations.

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